

RELENZA

Zanamivir

QUALITATIVE AND QUANTITATIVE COMPOSITION

Each *RELENZA ROTADISK* consists of four regularly spaced double foil blisters each containing a white to off-white micronised powder mixture of zanamivir (5 mg) and lactose (20 mg).

PHARMACEUTICAL FORM

Inhalation powder.

CLINICAL PARTICULARS

Indications

Treatment of Influenza: *RELENZA* is indicated for treatment of both influenza A and B in adults and children (5 years and older).

Prophylaxis: *RELENZA* is indicated for prophylaxis of both influenza A and B in adults and children (5 years and older).

Vaccination remains the primary method of preventing and controlling influenza.

RELENZA is indicated for prophylaxis of influenza A and B during community outbreaks only in circumstances where such prophylaxis is justified (such as when vaccine that antigenically matches circulating influenza is not available or there is a pandemic).
It is not recommended for routine prophylaxis against influenza infection.

Dosage and Administration

RELENZA is for administration to the respiratory tract by oral inhalation only, using the *DISKHALER* device provided.

Patients scheduled to take inhaled drugs, e.g. fast acting bronchodilators, at the same time as *RELENZA* should be advised to administer that drug prior to administration of *RELENZA*.

- *Adults*

Treatment of influenza

The recommended dose of *RELENZA* is two inhalations (2 x 5 mg) twice daily for five days, providing a total daily inhaled dose of 20mg.

For maximum benefit, treatment should begin as soon as possible (preferably within two days) after onset of symptoms.

Prophylaxis of influenza

The recommended dose of *RELENZA* is two inhalations (2 x 5 mg) once daily for 10 days, providing a total daily inhaled dose of 10 mg. This may be increased up to 28 days if the period of exposure risk extends beyond 10 days.

The full course of prophylaxis therapy should be completed as prescribed.

- *Children*

No dose modification is required (see *Pharmacokinetics*).

- *Elderly*

No dose modification is required (see *Pharmacokinetics*).

- *Renal impairment*

No dose modification is required (see *Pharmacokinetics*).

- *Hepatic impairment*

No dose modification is required (see *Pharmacokinetics*).

Contraindications

Hypersensitivity to any ingredient of the preparation (see *Pharmaceutical Particulars – List of Excipients*).

Contraindicated in patients with severe milk protein allergy.

Warnings and Precautions

Zanamivir is a specific treatment for infections due to Influenza A or B viruses. Use of zanamivir should be limited to patients who have characteristic symptoms of influenza when Influenza A or B virus infections have been documented locally.

Treatment of influenza in patients with severe asthma or other severe chronic respiratory diseases with zanamivir has not been adequately assessed due to the limited number of patients studied. In a placebo controlled study in patients with predominantly mild/moderate asthma and/or Chronic Obstructive Pulmonary Disease there was no evidence of a difference between zanamivir and placebo in forced expiratory volume in one second (FEV₁) or peak expiratory flow rate (PEFR) measured after the end of treatment. For influenza positive patients, there were small differences in favour of zanamivir in mean morning PEFR (12.9 L/min [95% CI 3.0 to 22.9 L/min] p=0.011), and in mean evening PEFR (13.1 L/min [95% CI 3.6 to 22.7 L/min] p=0.007).

Influenza infection can be associated with increased airways hyper-responsiveness. There have been very rare reports of patients being treated for influenza who have experienced bronchospasm and/or decline in respiratory function after the use of *RELENZA*, some of whom did not have any previous history of respiratory disease. Any such patients should discontinue *RELENZA* and seek medical evaluation. Patients with underlying respiratory disease should have a fast acting bronchodilator available when taking *RELENZA* (see *Dosage and Administration*).

RELENZA inhalation powder must not be made into an extemporaneous solution for administration by nebulisation or mechanical ventilation. There have been reports of hospitalised patients with influenza who received a solution made with *RELENZA* inhalation powder administered by nebulisation or mechanical ventilation, including a fatal case where it was reported that the lactose in this formulation obstructed the proper functioning of the equipment. *RELENZA* inhalation powder must only be administered using the device provided (see *Dosage and Administration*).

Influenza can be associated with a variety of neurological and behavioural symptoms. There have been postmarketing reports (mostly from Japan and in paediatric subjects) of seizures, delirium, hallucination and abnormal behavior in patients with influenza who were receiving neuraminidase inhibitors, including *RELENZA*. The events were observed mainly early in the illness and often had an abrupt onset and rapid resolution. The contribution of *RELENZA* to these events has not been established. If neuropsychiatric symptoms occur, the risks and benefits of continuing treatment should be evaluated for each patient.

Interactions

In vitro, zanamivir is not a substrate of cytochrome P450 (CYP) enzymes, P-glycoprotein (Pgp) or renal transporters nor does it affect human transporters (organic anion, cation, or urate transporters) or cytochrome P450 (CYP) enzymes (CYP1A2, 2A6, 2C9, 2C19, 2D6, 2E1 and 3A4). *In vivo*, zanamivir is excreted in urine as unchanged drug and there is no evidence that zanamivir is hepatically metabolized or modified.

Clinically significant drug interactions are unlikely.

Pregnancy and Lactation

Fertility

Animal studies indicate no clinically meaningful effects of zanamivir on male or female fertility (see *Pre-clinical Safety Data*).

Pregnancy

There are insufficient data on the use of *RELENZA* in pregnant women to inform drug-associated risk. Data from several studies have not found an increased risk of adverse pregnancy outcomes following *in utero* exposure to *RELENZA*, but due to limited sample sizes, no definitive conclusions can be drawn regarding the safety of *RELENZA* in pregnancy.

Reproductive studies performed in rats and rabbits indicated that placental transfer of zanamivir occurs and there was no evidence of teratogenicity. Results from a rat peri- and postnatal study showed no clinically meaningful impairment of offspring development. However, there is no information on placental transfer in humans.

As experience is limited, the use of zanamivir in pregnancy should be considered only if possible benefit to the patient is thought to outweigh any possible risk to the foetus.

Lactation

In rats zanamivir has been shown to be secreted in low amounts into milk. However, there is no information on secretion into breast milk in humans.

As experience is limited, the use of *RELENZA* in nursing mothers should be considered only if the possible benefit to the mother is thought to outweigh any possible risk to the child.

Effects on Ability to Drive and Use Machines

None known.

Adverse Reactions

Clinical trial data

In clinical studies, including those studies with high risk patients (the elderly, and patients with certain chronic medical conditions), the adverse events reported were similar in the *RELENZA* and placebo groups.

Table 1 lists common adverse events occurring at an incidence of $\geq 2\%$ reported during treatment and post-treatment in 5 major placebo-controlled influenza treatment clinical trials in adults, irrespective of the investigator's assessment of the possible relationship to study drug.

Table 1. Percentage of Patients Reporting Common Adverse Events (Incidence $\geq 2\%$): Major Treatment Studies in adults, irrespective of causality

Adverse Event	During Treatment		Post-treatment	
	Placebo (n=973)	Zanamivir (n=1703)	Placebo (n=973)	Zanamivir (n=1703)
Any event	39%	35%	26%	24%
Ear, nose and throat	4%	3%	3%	3%

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Nasal signs and symptoms	2%	2%	2%	2%
Throat and tonsil discomfort and pain	2%	2%	2%	2%
Ear, nose and throat infections				
Gastrointestinal				
Nausea and vomiting	4%	4%	1%	1%
Diarrhoea	3%	2%	1%	1%
Lower respiratory				
Cough	3%	2%	3%	2%
Bronchitis	3%	1%	2%	1%
Neurology				
Headaches	3%	3%	4%	4%

The percentage of patients reporting adverse events and the type during-treatment and post-treatment adverse events was similar for zanamivir and placebo groups. Most common adverse events were indistinguishable from signs and symptoms of influenza-like illness. Diarrhoea, dizziness, nausea and vomiting have been reported but there was no clear causal association with study treatment in the adult studies.

The nature and frequency of reports of adverse events in children (Study NAI30009) was similar to that reported in adults. 2% of both placebo and zanamivir recipients respectively reported adverse events thought to be drug-related.

Prophylaxis studies

Adverse events that occurred with an incidence of $\geq 1.5\%$ in the 4 prophylaxis studies are listed in Table 2.

Table 2. Summary of Adverse Events $\geq 1.5\%$ Incidence During Prophylaxis Studies in Adults, Adolescents, and Children*

Adverse Event	Household Studies NAI30010, NAI30031		Community Studies NAIA3005, NAI30034	
	RELENZA contacts (n = 1,068)	Placebo [†] contacts (n = 1,059)	RELENZA (n = 2,231)	Placebo [†] (n = 2,239)
Lower respiratory				
Viral respiratory infections	13%	19%	3%	4%
Cough	7%	9%	17%	18%
Bronchospasm-like events	<1%	1%	2%	3%
Neurologic				
Headaches	13%	14%	24%	26%
Ear, nose, and throat				
Nasal signs and symptoms	12%	12%	12%	13%
Throat and tonsil discomfort and pain	8%	9%	19%	20%
Nasal inflammation	1%	2%		
Ear, nose, throat infections			2%	2%
Musculoskeletal				
Muscle pain	3%	3%	8%	8%
Musculoskeletal pain			6%	6%
Arthralgia and articular rheumatism			2%	<1%

Endocrine and metabolic				
Feeding problems (decreased or increased appetite and anorexia)	2%	2%	4%	4%
Gastrointestinal				
Nausea and vomiting	1%	2%	2%	3%
Diarrhoea			2%	2%
Non-site specific				
Malaise and fatigue	5%	5%	8%	8%
Temperature regulation disturbances (fever and/or chills)	5%	4%	9%	10%

* In prophylaxis studies symptoms associated with influenza-like illness were captured as adverse events; subjects were enrolled during a winter respiratory season during which time any symptoms that occurred were captured as adverse events.

† Because the placebo consisted of inhaled lactose powder, which is also the vehicle for the active drug, some adverse events occurring at similar frequencies in different treatment groups could be related to lactose vehicle inhalation.

The frequency and nature of adverse events reported during prophylaxis for all four studies were very similar between the placebo and zanamivir groups. The most commonly reported adverse events during prophylaxis were headaches, throat and tonsil discomfort and pain, cough, and nasal signs and symptoms, which are typical of the signs and symptoms of influenza and other viral respiratory infections.

There were no notable differences in the frequency and nature of adverse events between placebo and zanamivir in paediatric subgroups (ages 5 to 11 and 12 to 16 years).

High Risk Population

In a treatment study in asthma and/or COPD patients (NAI30008), the proportion of patients reporting adverse events that were thought to be related to treatment was the same for both zanamivir and placebo recipients (9%).

The proportion of high risk patients reporting drug-related adverse events was the same for both zanamivir and placebo recipients (10%) when the data was combined for 8 major placebo-controlled influenza treatment clinical trials. There were 7 studies in adult, including NAI30008, and one study in paediatric patients.

Inhaled zanamivir had an acceptable safety profile in prophylactic use in high risk subjects in Study NAI30034. In general, the frequency and nature of adverse events was similar across treatment groups for subjects with each category of underlying high-risk condition.

Post-marketing data

The following reactions have been identified during post-approval use of *RELENZA* for the treatment of influenza.

Immune Systems Disorders:

Very rare: Allergic-type reactions, including anaphylactic and anaphylactoid reactions, facial and oropharyngeal oedema.

Nervous systems disorders:

Very rare: Vasovagal-like reactions have been reported in patients with influenza symptoms, such as fever and dehydration, shortly following inhalation of *RELENZA*.

Respiratory, thoracic and mediastinal disorders:

Very rare: Bronchospasm, dyspnoea.

Skin and subcutaneous tissue disorders:

Very rare: Rash, urticaria.

Very rare: Severe skin reactions including Erythema Multiforme, Stevens-Johnson syndrome, Toxic epidermal necrolysis

Overdose

Symptoms and Signs

Reports of overdoses with *RELENZA* have been received during postmarketing experience. The reported clinical signs or symptoms were similar to those observed with therapeutic doses of *RELENZA* and the underlying disease.

Doses of an investigational (lactose-free) aqueous solution of zanamivir up to 96 mg/day (approximately 5 times the maximum daily recommended dose) have been administered intranasally without adverse effects. Additionally, systemic exposure by intravenous administration of up to 1200 mg/day for five days showed no adverse effect.

Treatment

As zanamivir has a low molecular weight, low protein binding, and small volume of distribution, it is expected to be removed by haemodialysis. Further management should be as clinically indicated or as recommended by the national poisons centre, where available.

PHARMACOLOGICAL PROPERTIES

Pharmacodynamics

ATC Code

J05AH01

Mechanism of Action

Zanamivir is a potent and highly selective inhibitor of neuraminidase, the influenza virus surface enzyme. Viral neuraminidase aids the release of newly formed virus particles from infected cells, and may facilitate access of virus through mucus to epithelial cell surfaces, to allow viral infection of other cells. The inhibition of this enzyme is reflected in both *in vitro* and *in vivo* activity against influenza A and B virus replication, and encompasses all of the known neuraminidase subtypes of influenza A viruses.

The activity of zanamivir is extracellular. It reduces the propagation of both influenza A and B viruses by inhibiting the release of infectious influenza virions from the epithelial cells of the respiratory tract. Influenza viral replication is confined to the superficial epithelium of the respiratory tract. The efficacy of topical administration of zanamivir to this site has been confirmed in clinical studies. Clinical trial data have shown that treatment of acute influenza infections with *RELENZA* produces reductions in virus shedding from the respiratory tract compared to placebo. Emergence of virus with reduced susceptibility to zanamivir in the clinical trials of zanamivir was rare.

When taken as recommended for the treatment of influenza, zanamivir alleviates and reduces duration of influenza symptoms in patients aged 5 years and older. The efficacy of zanamivir is optimal if treatment is initiated as early as possible after symptom onset. The efficacy of zanamivir administered more than 48 hours after the onset of symptoms has not been demonstrated. When taken as recommended for the prophylaxis of influenza, it reduces the incidence of symptomatic influenza in the same population.

Drug Resistance

Influenza viruses with reduced susceptibility to zanamivir have been recovered *in vitro* by passage of the virus in the presence of increasing concentrations of the drug. Genetic analysis of these viruses showed that the reduced susceptibility *in vitro* to zanamivir is associated with mutations that result in amino acid changes in the viral neuraminidase or viral haemagglutinin or both. Even though exhibiting a drug resistant phenotype *in vitro*, based on *in vivo* studies to date, a virus with such combinations of mutations would be expected to have reduced pathogenicity and transmissibility in the clinical setting.

There has been no detectable emergence of virus to date with reduced susceptibility to zanamivir during the clinical development programme. Data obtained from *in vitro* studies and from clinical treatment and prophylaxis studies suggests that the potential for the development of reduced clinical susceptibility to zanamivir in the future is low.

Pharmacokinetic Properties

Absorption

Pharmacokinetic studies in man following inhalation of zanamivir suggest that approximately 4 - 17% of the dose is absorbed into the systemic circulation. Serum zanamivir concentrations generally peak within 1-2 hours.

Distribution

After oral inhalation, zanamivir is widely deposited in the respiratory tract mucosa, thus delivering the drug to the site of influenza infection. The two major sites of deposition are the oropharynx and the lungs (mean 77.6% and 13.2%, respectively). Following twice daily administration of zanamivir 10 mg by oral inhalation, the median trough concentrations of zanamivir measured at the epithelial layer of the airways (the major sites of influenza viral replication) ranged from 326 ng/mL to 891 ng/mL. These trough concentrations are multiple-fold in excess of the *in vitro* IC₅₀ (<1 to 2.6 ng/mL) values for influenza virus neuraminidase for various influenza A (N1 and N2 subtypes) and B viruses. , Despite a large proportion of the inhaled dose being swallowed, the absolute oral bioavailability is low (about 2%) and swallowed drug does not contribute significantly to systemic exposure. As an index of systemic absorption, between 3.8% to 17% (median values) of the inhaled dose as dry powder via DISKHALER was excreted unchanged in the urine. There is no evidence of modification in kinetics after repeated dosing with inhaled administration. The plasma protein binding of zanamivir is low, less than 10%, as shown by *in vitro* assay in the presence of 0.5 and 10 micrograms/mL zanamivir.

Metabolism

Animal and human studies show that zanamivir is excreted as unchanged drug. Chromatographic profiling of urine samples from rats, dogs and rabbits show no evidence of biotransformation. These findings indicate that zanamivir does not undergo metabolism.

Excretion

The median serum half-life of zanamivir following intravenous administration ranges from 1.5 to 3.1 hours. The delayed serum half-life following oral inhalation (range 2.5 to 5.1 hours) reflects slow systemic absorption by this route. Zanamivir is excreted unchanged in the urine. Total clearance ranges from 2.5 to 10.9 L/h as approximated by urinary clearance. Elimination is complete within 24 hours.

Pharmacokinetics in Renal Impairment

Dose modification is not required.

Drug clearance is proportional to creatinine clearance; a 50% reduction in renal clearance results in a corresponding 50% reduction in serum clearance of zanamivir. In a single dose study with intravenous zanamivir (2 mg or 4 mg), in patients with severe renal impairment (creatinine clearance <25 mL/min), systemic exposure to zanamivir was increased approximately 7 fold relative to patients with normal renal function. However, oral inhalation results in very low systemic exposure (approximately 4 to 17% of the dose administered) and in studies much higher systemic exposure (600 mg IV for 5 days) was well tolerated.

Pharmacokinetics in Paediatric Patients

In an open-label single-dose study the pharmacokinetics of zanamivir have been evaluated in twenty-four children aged 3 months to 12 years using nebulised (10mg) and dry powder (10mg) inhalation formulations. Six patients had either undetectable zanamivir concentrations or had low drug concentrations after 1.5 hours. Eleven patients, aged 5 to 12 years, administered the dry powder formulation had C_{max} median values of 43 ng/mL (range 15 to 74) and AUC_∞ median values of 167 ng.hour/mL (range 58 to 279).

Clinical Studies

Treatment

Five Phase III randomised, placebo-controlled, parallel-group, multicentre studies were conducted with zanamivir in the treatment of naturally acquired influenza A and B in adults and adolescents (NAIB3001, NAIA3002, NAIB3002 and NAI30008) and in children aged 5-12 years (NAI30009). The Intent-to-Treat population of the four adult studies and the paediatric study comprised of 2113 and 471 patients respectively; of whom 1075 adults and 224 children received 10mg zanamivir twice daily by oral inhalation. The proportion of influenza-positive patients across the five studies, ranges from 60% to 78%. Of these, 88-91% had influenza A and 9-11% had influenza B in the adult studies. In the paediatric study, there were 65% and 35% influenza A and B cases respectively.

In the adult studies, patients aged 65 years and over, or patients with concurrent cardiovascular conditions (excluding uncomplicated hypertension), respiratory conditions, metabolic conditions, endocrine conditions or who were immunocompromised were considered more vulnerable to complications of influenza and were therefore categorised as 'High Risk' patients.

Protocols for the five studies were similar. Patients were recruited into studies when influenza was known to be circulating in the community. Eligibility of entry into study was based on the presence of feverishness (NAIB3001) or fever (NAIA3002, NAIB3002 and NAI30008) and at least two of the following: headache, myalgia, cough and sore throat. For the paediatric study (NAI30009), this was simplified to fever, with no clinical evidence of a bacterial infection. Where fever was a criterion, the requirement was temperature $\geq 37.8^{\circ}\text{C}$ except in patients aged 65 years or older ($\geq 37.2^{\circ}\text{C}$).

In NAIB3001 and NAI30009, patients were recruited and commenced using study medication within 36 hours of the onset of symptoms, while in NAIA3002 and NAIB3002, patients were recruited and commenced treatment within two calendar days.

The primary endpoint was identical for all four Phase III studies, i.e. time to alleviation of clinically significant signs and symptoms of influenza. Alleviation was defined as no fever (temperature $<37.8^{\circ}\text{C}$ and feverishness score of 'none') and headache, myalgia, cough and sore throat recorded as 'none' or 'mild' and maintained for 24 hours.

Results of primary endpoint analyses for the Intent-to-Treat and Influenza Positive populations are shown below.

Table 3. Median Time (days) to Alleviation of Clinically Significant Influenza Symptoms in Phase III Studies

Study	Placebo (median day)	Zanamivir (median day)	Days Difference	95% Confidence Interval		p-value	Placebo N	Zanamivir N
Intent-to-Treat Population								
NAIB3001	6.5	5.0	1.5	0.5	2.25	0.011	228	227
NAIA3002	6	5.5	0.5	-0.5	1.0	0.228	365	412
NAIB3002	7.5	5.0	2.5	0.75	3.5	<0.001	182	174
NAI30008	7.0	6.0	1.0	0.0	2.0	0.123	263	262
NAI30009	5.0	4.5	0.5	0.0	1.5	0.011	247	224
Influenza Positive Population								
NAIB3001	6	4.5	1.5	0.5	2.5	0.004	160	161
NAIA3002	6	5	1.0	0.0	1.5	0.078	257	312
NAIB3002	7.5	5	2.5	1.0	4.0	<0.001	141	136
NAI30008	7.0	5.5	1.50	0.5	3.25	0.009	153	160
NAI30009	5.25	4.0	1.25	0.5	2.0	<0.001	182	164

The four adult studies included 361 'High Risk' patients treated with zanamivir. Of these, 58 were aged 65 years or older. In a pooled analysis of the Influenza Positive population in the principle phase III treatment studies (NAIB3001, NAIA3002, NAIB3002 and NAI30008) the median time to alleviation of influenza symptoms was reduced by 1.5 days for patients taking *RELENZA* as compared to placebo ($p<0.001$). Complications were reduced from 208/711 (29%) of placebo patients to 171/769 (22%) of zanamivir

patients (relative risk: 0.77; 95% CI: 0.65 to 0.92; p=0.004). Use of antibiotics for treatment of complications was reduced from 136/711 (19%) of placebo patients to 110/769 (14%) of zanamivir patients (relative risk: 0.76; 95% CI: 0.60 to 0.95; p=0.021).

In the paediatric study, median time to symptom improvement was one day shorter in patients receiving zanamivir compared with placebo. Comparable rates of development of complications were observed between the treatment groups.

Study NAI30008 enrolled patients aged 12 years and over, diagnosed with asthma or chronic obstructive pulmonary disease. Baseline status for influenza positive patients is shown in Table 4.

Table 4. Baseline status for influenza positive patients in Study NAI30008

	Placebo (n=153)	Zanamivir (n=160)
Asthma and COPD	8 (5%)	8 (5%)
Asthma only	120 (78%)	127 (79%)
COPD only	23 (15%)	25 (16%)
Neither*	2 (1%)	0

*Subjects had asthma/COPD in the opinion of the investigator but did not have a documented history.

The median time to symptom improvement was 1.5 days less in patients receiving zanamivir compared with placebo. Comparable rates of development of complications were observed between the treatment groups (see *Warnings and Precautions*).

There have been no studies of treatment of influenza, where treatment was commenced after 48 hours of onset of symptoms.

Prophylaxis

The efficacy of *RELENZA* in preventing naturally occurring influenza illness has been demonstrated in two post-exposure prophylaxis studies in households (NAIA30010 and NAI30031) and two seasonal prophylaxis studies during community outbreaks of influenza (NAIA3005 and NAI30034). The primary efficacy endpoint in these studies was the incidence of symptomatic, laboratory-confirmed influenza, defined as the presence of two or more of the following symptoms: oral temperature 37.8C or feverishness, cough, headache, sore throat, and myalgia; and laboratory confirmation of influenza by culture, PCR, or seroconversion (defined as a 4-fold increase in convalescent antibody titer from baseline).

Two studies assessed post-exposure prophylaxis in household contacts once a member of the household (the index case) developed an influenza-like illness. Within 1.5 days of onset of symptoms in an index case, each household (including all family members 5 years of age) was randomized to *RELENZA* 10 mg inhaled once daily or placebo inhaled once daily for 10 days. In the first study (NAIA30010) only, each index case was randomized to the same treatment (*RELENZA* or placebo) as the other household members. In this study, the proportion of households with at least one new case of symptomatic influenza was reduced from 19% (32 of 168 households) with placebo to 4% (7 of 169 households) with *RELENZA* (79% protective efficacy). In the second study (NAI30031), index cases were provided with relief medication for supportive care and the incidence of symptomatic influenza was reduced from 19% (46 of 242 households) with placebo to 4% (10 of 245 households) with *RELENZA* (81% protective efficacy). Results were similar in the subgroups with influenza A or B.

Two seasonal prophylaxis studies assessed *RELENZA* 10 mg inhaled once daily versus placebo inhaled once daily for 28 days during community outbreaks. In study NAIA3005, patients had to be part of a university community during an influenza outbreak, be aged 18 years or older and enter the study within 72 hours of an outbreak being declared. In this study the incidence of symptomatic influenza was reduced from 6.1% (34 of 554) with placebo to 2.0% (11 of 553) with *RELENZA* (67% protective efficacy).

In the second study (NAI30034), community-dwelling subjects, 12 years and older, who were at high risk of complications from influenza were enrolled. High risk was defined as subjects >65 years of age, subjects with diabetes mellitus and subjects with chronic disorders of the pulmonary or cardiovascular systems. The study was conducted during a season with low influenza activity. Despite this, the study demonstrated statistically significant protective benefit in subjects who received *RELENZA*. The incidence of symptomatic influenza was reduced from 1.4% (23 of 1,685) with placebo to 0.2% (4 of 1,678) with *RELENZA* (83% protective efficacy).

Results of the primary efficacy endpoint analysis are presented for the Intent To Treat population for each of the studies are shown below.

Table 5 Summary of Primary Efficacy Analyses: Primary Studies (ITT Population)

Study	Cases of Influenza		p-value	Approximate Relative Risk (95% CI)	Protective Efficacy (%)
	Placebo n/N (%)	Zanamivir n/N (%)			
NAI30010 ¹	32/168 (19)	7/169 (4)	<0.001	0.21 (0.11, 0.43)	79
NAI30031 ¹	46/242 (19)	10/245 (4)	<0.001	0.19 (0.10, 0.36)	81
NAIA3005 ²	34/554 (6)	11/553 (2)	<0.001	0.33 (0.17, 0.61)	67
NAI30034	23/1685 (1)	4/1678 (<1)	<0.001	0.17 (0.07, 0.44)	83

1. In at least one contact case in the family/household

2. In all subjects (not just non-vaccinated subjects)

Table 6 Summary of Primary Efficacy Analyses by age in pivotal studies (ITT Population)

Study	Age group (years)	Contact cases/subjects with symptomatic laboratory confirmed influenza	
		Placebo n/N (%)	Zanamivir n/N (%)
Family/Household Studies			
NAI30010	5 - 6	4/33 (12)	1/28 (4)
	7 - 11	7/91 (8)	2/89 (2)
	12 - 16	2/64 (3)	2/65 (3)
	17 – 34	4/53 (8)	1/58 (2)
	35 – 49	21/160 (13)	1/151 (<1)
	50 – 64	1/20 (5)	0/23
	65+	1/2 (50)	0/0
NAI30031	5 - 6	4/26 (15)	1/36 (3)
	7 - 11	15/128 (12)	6/122 (5)
	12 - 16	9/100 (9)	0/124
	17 – 34	6/100 (6)	3/91 (3)
	35 – 49	20/240 (8)	2/250 (<1)
	50 – 64	1/32 (3)	0/31
	65+	0/4	0/7
Community Studies			
NAIA3005	17 – 34	27/416 (6)	9/416 (2)
	35 – 49	5/116 (4)	2/105 (2)
	50 – 64	2/21 (10)	0/31
	65+	0/1	0/1
NAI30034	12- 16	3/55 (5)	1/51 (2)
	17 – 34	4/114 (4)	1/123 (<1)
	35 – 49	7/246 (3)	1/128 (<1)
	50 – 64	4/320 (1)	0/330

	65+	5/950 (<1)	1/946 (<1)
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Two prophylaxis studies conducted in a nursing home setting assessed *RELENZA* 10 mg once daily for 14 days compared with standard of care in the prevention of influenza infection. In study NAIA3003 standard of care was rimantadine for influenza A and placebo for influenza B. In study NAIA3004 standard of care was placebo. There was no upper age limit, and once an influenza outbreak was declared in the nursing home, study treatment was initiated. The primary endpoint for both studies was the proportion of subjects who during prophylaxis, developed a new sign or symptom and had laboratory confirmation of influenza. The results of Study NAIA3003 showed a trend toward efficacy against an active comparator, although the results did not reach statistical significance (protective efficacy 56%, p-value 0.085). The results of Study NAIA3004 did not demonstrate significant protective efficacy (protective efficacy 29%, p-value 0.355).

Pre-clinical Safety Data

Administration of zanamivir in animal toxicity studies was not associated with any clinically relevant effects. Zanamivir was not genotoxic and showed no evidence of carcinogenic potential in long term carcinogenicity studies in rats and mice.

No drug-related malformations, maternal toxicity or embryotoxicity were observed in pregnant rats or rabbits or their fetuses following intravenous administration of zanamivir at doses up to 90 mg/kg/day. Following subcutaneous administration of zanamivir in an additional rat embryofoetal development study, there was an increase in the incidence rates of a variety of minor skeletal and visceral alterations and variants in the exposed offspring at the highest dose 80 mg/kg, three times daily (240 mg/kg/day; total daily dose), most of which remained within the background rates of the historical occurrence in the strain studied. Based on AUC measurements, the 80 mg/kg dose (240 mg/kg/day) produced an exposure approximately 3 or 1000 times the human exposure at the clinical intravenous or inhaled dose, respectively. In the peri- and post-natal developmental study conducted in rats, there was no clinically meaningful impairment of development of offspring.

Intravenous doses of up to 90mg/kg/day zanamivir produced no effect on fertility and reproductive function of the treated or subsequent generation in male and female rats.

PHARMACEUTICAL PARTICULARS

List of Excipients

Lactose (which contains milk protein).

Incompatibilities

None known

Shelf Life

The expiry date is indicated on the packaging.

Special Precautions for Storage

RELENZA ROTADISKS should not be stored above 30°C.

Nature and Contents of Containers

RELENZA ROTADISKS consist of a circular foil disk (a *ROTADISK*) with four regularly distributed blisters each containing 5 mg of zanamivir and 20 mg of lactose. A *DISKHALER* is provided to administer the medication.

Instructions for Use/Handling

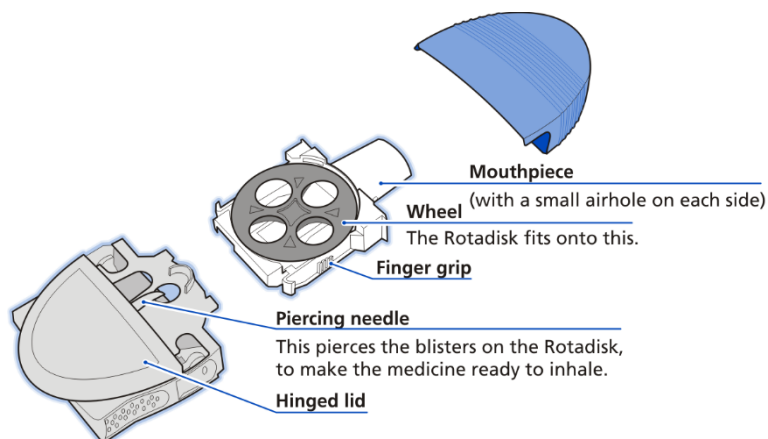
The powdered medicine is inhaled through the mouth into the lungs. The *DISKHALER* device is loaded with a disk which contains the medicine in individual blisters which are opened as the device is manipulated.

Instructions for the Patient:

Read the step-by-step guide carefully before you use your first dose. If you're still not sure how to use the *DISKHALER*, ask your pharmacist to go through the instructions with you.

The *DISKHALER* has three parts:

Don't take it apart until you have looked at the step-by-step guide.



The *ROTADISK* fits into the *DISKHALER*



The *ROTADISK* fits onto the wheel of the *DISKHALER*.

The *ROTADISK* contains four blisters. Each blister contains 5 mg of *RELENZA*. A dose is normally two blisters (10 mg).

Important:

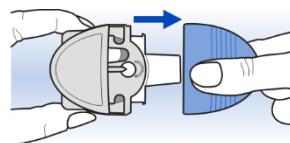
- Don't pierce any of the blisters on the *ROTADISK* before you load it onto the *DISKHALER*.
- You can keep a *ROTADISK* on the *DISKHALER* between doses, but don't pierce a blister until just before you inhale your dose.
- Keep the *DISKHALER* clean. Wipe the mouthpiece with a tissue after you use it, and replace the blue cover between uses.

STEP-BY-STEP GUIDE TO USING YOUR RELENZA DISKHALER

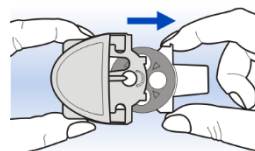
To load a *ROTADISK* into the *DISKHALER*:

1. Remove the blue cover.

Check that the mouthpiece is clean, inside and outside.



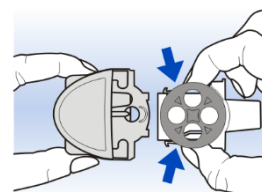
2. Hold the white sliding tray as shown and pull it out until it stops.



3. Gently squeeze the finger grips on the sides of the white tray.

Remove the tray from the main body.

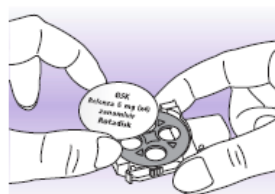
The white tray should come out easily.



4. Place a new *RELENZA ROTADISK* on the wheel.

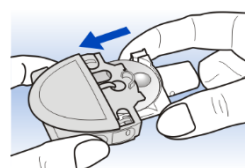
Make sure the printed side is up, with the blisters facing downwards.

The blisters fit into the holes in the wheel.



5. Push the white tray back into the main body.

If you're not ready to inhale a dose of *RELENZA* straight away, replace the blue cover.



To get your dose ready to inhale:

Don't do this until just before you inhale a dose.

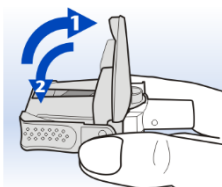
6. Hold the *DISKHALER* horizontally.

Flip the lid up as far as it will go.

The lid must be fully vertical, to make sure that the blister is pierced at both the top and bottom.

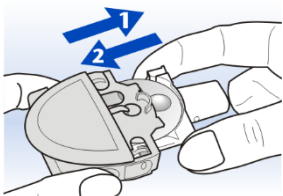
Push the lid back down.

Your *DISKHALER* is now ready for use. Keep it horizontal until you have inhaled your dose.



Keep the *DISKHALER* horizontal

To inhale the medication:

7. Sit down in a comfortable position. Don't put the <i>DISKHALER</i> into your mouth yet. Breathe out as far as is comfortable, keeping the <i>DISKHALER</i> away from your mouth. Don't blow into the <i>DISKHALER</i>. If you do, you'll blow the powder out of the <i>ROTADISK</i>. Place the mouthpiece between your teeth. Close your lips firmly around the mouthpiece. Don't bite the mouthpiece. Don't block the airholes on the side of the mouthpiece. Take one quick, deep breath in through the mouthpiece. Hold this breath for a few seconds or as long as is comfortable.	 Keep the <i>DISKHALER</i> horizontal
To prepare the next blister (the second part of your dose):	
8. Pull the white tray out as far as it will go (don't remove it completely), then push it back in again. This will turn the wheel so the next blister will appear. Repeat if necessary until a full blister is positioned under the piercing needle. Repeat steps 6 and 7 to inhale the medicine.	
9. After you've inhaled the full dose (normally two blisters): Wipe the mouthpiece with a tissue and replace the blue cover. It's important to keep the <i>DISKHALER</i> clean.	
To replace the <i>ROTADISK</i>:	
10. When all four blisters are empty, remove the <i>ROTADISK</i> from the <i>DISKHALER</i> and insert a new one, using steps 1 to 5.	

Not all presentations are available in every country.

Manufactured by:

GlaxoSmithKline Australia Pty Ltd. Boronia, Australia

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